

Hand Gesture Recognition Using EMG Sensors for Command Control in Robotic Interfaces.

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Abstract

This work presents a low-cost, single-channel electromyographic (EMG) signal acquisition and processing system oriented towards hand gesture recognition for human-machine interaction applications and control signal generation. The signal was acquired using a surface sensor connected to an Arduino Uno at a sampling frequency of 1000 Hz. A digital processing and classification chain based on Random Forest was developed for automatic muscle gesture recognition. The system was evaluated on three hand gestures (closed fist, thumb up and open palm) and the rest state.

1. Introduction

Interfaces based on electromyographic (EMG) signals have gained great relevance in applications such as myoelectric prosthetics, robotics and human-machine interaction systems, due to their capacity to capture muscle activity in a non-invasive way.

Works such as those by Benalcázar [1] have addressed gesture classification using EMG signals, highlighting the use of multi-channel configurations that allow capturing spatial information of muscle activity and improving model performance. In this context, public databases such as NinaPro [2], which provide high-quality multi-channel EMG signals, have been widely used as a standard for training and evaluating classification algorithms.

However, multi-channel systems present higher computational and acquisition costs, which makes their use difficult in portable or low-cost applications. For this reason, this work proposes a single-channel EMG acquisition and processing system oriented towards hand gesture recognition.

This work focuses exclusively on hand gesture recognition using electromyographic (EMG) signals for possible use in gestural control systems applied to robotic interfaces. The scope of the study includes EMG signal acquisition, digital processing, muscle activation segmentation, feature extraction and automatic gesture classification using machine learning techniques.

Furthermore, the system includes a basic command interpretation and execution architecture oriented towards gestural control applications. First, the Random Forest classifier processes the features extracted from the EMG signal and generates a numerical label associated with the detected gesture. This label is temporarily stored in a shared memory buffer that acts as an interface between the classification stage and the actuation layer. Subsequently,

a second independent process continuously monitors this buffer and translates the received label into a specific control instruction for the corresponding actuator or robotic system. Finally, once the physical action associated with the gesture has been executed, the buffer content is automatically deleted in order to avoid repeated executions of the same command in subsequent cycles. This decoupled architecture allows separating the electromyographic processing from the physical actuation layer, facilitating scalability and future integration on real robotic hardware, although this definitive physical implementation is outside the scope of the present work.

2. Methodology

The EMG signal was acquired using a surface sensor (SEN0240, DFRobot) connected to an Arduino Uno, at a sampling frequency of 1000 Hz. The electrodes were placed on the forearm flexor muscle, approximately 4–5 cm from the medial epicondyle. Conductive gel was used to reduce skin-electrode impedance and improve signal quality, thus increasing the signal-to-noise ratio (SNR).

Signal pre-processing consisted of applying a Notch filter at 50 Hz to eliminate powerline interference, followed by a bandpass filter between 20 and 450 Hz, corresponding to the characteristic physiological range of surface EMG signals [5]. Subsequently, full-wave rectification of the signal was performed and smoothing was applied using a 50 ms moving average, thus obtaining the signal envelope.

A 50 ms window was used as it provides an adequate trade-off between smoothing and preservation of the temporal dynamics of the signal.

Validation of the filtering process is shown in Figures 1, 2 and 3, where the elimination of the 50 Hz component and the limitation of the spectrum to the physiological range can be observed.

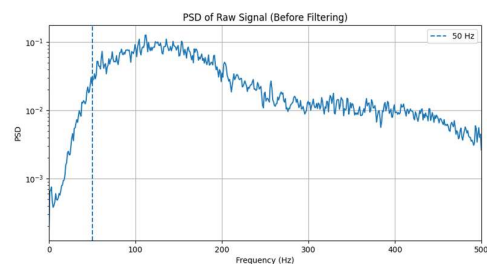


Figure 1. PSD raw

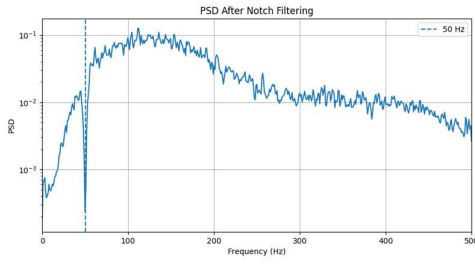


Figure 2. PSD after Notch filter

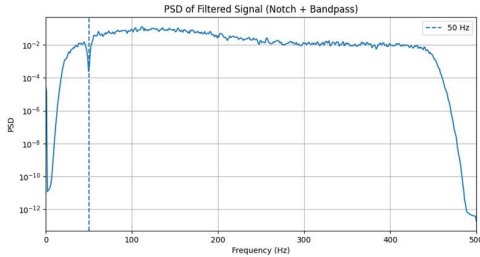


Figure 3. PSD after band-pass filter (20-450 Hz)

The power spectral density (PSD) analysis shows a clear reduction of the 50 Hz component after filtering, which validates the effectiveness of the pre-processing applied.

Subsequently, the application of the bandpass filter (20–450 Hz) eliminates the low-frequency components associated with motion artefacts, as well as high-frequency noise, concentrating the signal energy in the characteristic physiological range of surface EMG.

Activation detection was performed using an adaptive threshold defined on the envelope. It was observed that low values of α (sensitivity factor) increased the sensitivity of the system, generating spurious detections due to background noise, while high values caused the loss of real activations. The final selected value (0.3) allowed an adequate trade-off between sensitivity and robustness, enabling consistent detection of muscle activity regions without introducing excessive fragmentation or isolated false activations. Furthermore, this value produced temporally more stable binary masks that were more coherent with the visual morphology of the processed EMG signal, subsequently facilitating the reliable segmentation and feature extraction.

In order to improve robustness against noise and avoid isolated short-duration detections, a temporal consistency filter was applied using a 200 ms moving average on the obtained binary mask. This procedure allowed smoothing abrupt transitions and eliminating spurious activations generated by small residual signal fluctuations or by momentary threshold crossings. The 200 ms temporal window was selected experimentally as it provides an adequate balance between stability and temporal resolution; shorter windows maintained false detections of very brief duration, while excessively long windows tended to merge nearby activations and degrade temporal precision in the delimitation of muscle gestures.

Figure 4 shows the rectified signal, the smoothed envelope, and the applied adaptive threshold.

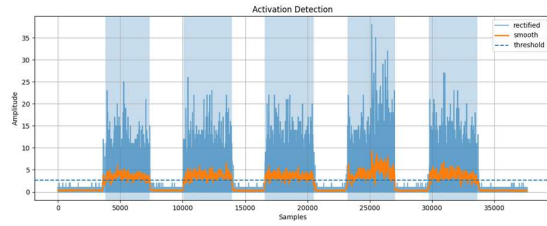


Figure 4. Segmentation using an adaptive threshold

The regularized mask is obtained through a second binarization, where $\theta = 0.3$ is the temporal persistence threshold used, which was chosen after analyzing its effect on the continuity of the detected segments. Lower values resulted in fragmented and noise-sensitive activations, while higher values eliminated valid segments. The selected value allowed retaining only activations with sufficient temporal persistence, improving the quality of the segments used in subsequent stages.

This procedure allows eliminating spurious short-duration activations and avoiding segment fragmentation. Additionally, a minimum duration criterion of 0.5 s was applied to discard non-physiological segments, as well as a segment merging process using a temporal threshold of 0.8 s. Finally, a 200 ms trimming was performed at the edges of each segment in order to eliminate the transition zones between rest and activation to avoid their influence on feature extraction.

From the obtained segments, temporal features were extracted using 200 ms sliding windows with 50% overlap, which allowed increasing the temporal resolution of the extracted features while maintaining moderate redundancy between consecutive windows. The features used were the root mean square (RMS), mean absolute value (MAV), zero crossing count (ZC), variance (VAR) and waveform length (WL) [4].

The signal-to-noise ratio (SNR) was estimated from the ratio between signal power in activation and rest windows, obtaining an SNR of around 10 dB across the three gestures. Power was calculated using the RMS value.

For the benchmarking process with the NinaPro database, the signal was subjected to the same digital processing chain. The repetitions were aligned with respect to the activation peak and trimmed using a 5-second temporal window centered on that peak, in order to guarantee comparability between signals. Subsequently, repetition averaging (folding) was performed, obtaining a mean signal representative of the gesture.

Finally, both the experimental signal and the reference signal were normalized to the range [0,1], allowing a direct comparison of their morphology.

The similarity between both signals was evaluated using the Pearson correlation coefficient and the root mean square error (RMSE), while the performance of the classification system was evaluated using a Random Forest model [3] with GroupKFold cross-validation with 5 partitions. To avoid information leakage between overlapping windows, all windows belonging to the same muscle contraction were grouped within the same group,

ensuring that no repetition appeared simultaneously in training and test.

3. Results and Benchmarking

In order to evaluate the morphological coherence of the acquired EMG signal, a benchmarking process was carried out with the NinaPro database. For this purpose, both the experimental signal and the reference signal were subjected to the same digital processing chain, including filtering, rectification and envelope extraction.

The morphological similarity between both signals was evaluated using the Pearson correlation coefficient. This coefficient takes values in the range $[-1, 1]$, with values close to 1 indicating high similarity between signals. Additionally, the p-value associated with the correlation coefficient was calculated, which allows evaluating the statistical significance of the observed relationship.

Additionally, the root mean square error (RMSE) was used as a complementary measure of the difference between both signals.

The results obtained show high morphological similarity between the experimental signal and the NinaPro reference signal, especially after the alignment and averaging (folding) process. The morphological comparisons between the experimental signal and the NinaPro reference signal for the different gestures are shown in Figures 5, 6 and 7.

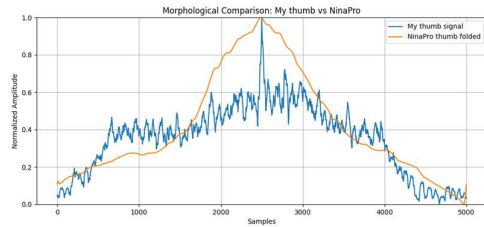


Figure 5. My thumb vs Ninapro

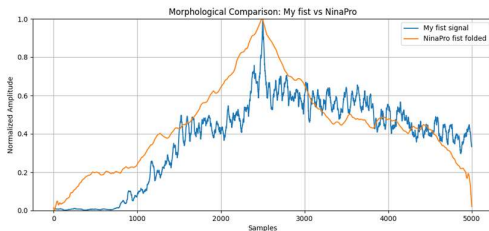


Figure 6. My fist vs Ninapro

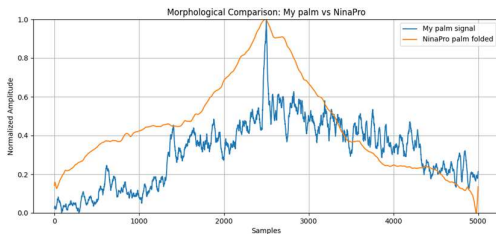


Figure 7. My palm vs Ninapro

The quantitative morphological similarity values obtained for each gesture are summarised in Table 1.

Gesture	Pearson (r)	p-value	RMSE
Thumb	0.85255	0.0	0.16743
Fist	0.81896	0.0	0.15163
Palm	0.65757	0.0	0.22506

Table 1. Quantitative similarity values

In particular, high values of the Pearson correlation coefficient were obtained, accompanied by significantly low p-values ($p < 0.05$), indicating that the observed similarity is not due to chance, but reflects a statistically significant relationship between both signals. The RMSE values obtained were low, reinforcing the agreement between the compared signals.

Regarding the performance of the classification system, a Random Forest model was used with the extracted temporal features (RMS, MAV, ZC, VAR and WL) as input variables. The evaluation was carried out using GroupKFold cross-validation with 5 partitions.

The per-class analysis showed precisions of 0.99 for the neutral state, 0.92 for the palm gesture, 0.90 for the thumb gesture and 0.85 for the fist gesture. The average per-class precision (macro-precision) obtained was 0.916, providing a more balanced estimate of model performance across the different gestures considered.

Although the Pearson correlation coefficient obtained for the palm gesture is lower than that of other gestures, the classification model maintains high performance. This result can be explained by the difference between overall morphological similarity and the discriminative capacity of the features used. While the Pearson coefficient evaluates similarity in the complete shape of the signal, the Random Forest model is based on statistical features such as RMS, MAV, VAR and WL, capable of capturing relevant energetic and temporal properties of the EMG signal.

Although the neutral state has a larger number of samples and therefore also a particularly high precision, the evaluation using macro-precision allowed obtaining a more balanced and honest estimate of classifier performance across the different gestures considered.

The resulting confusion matrix is shown in Figure 8, where a high degree of accuracy in the classification of the different gestures can be observed. However, certain confusions are detected mainly between the fist and thumb gestures, as well as between fist and palm, suggesting the existence of partially similar muscle activation patterns between these classes.

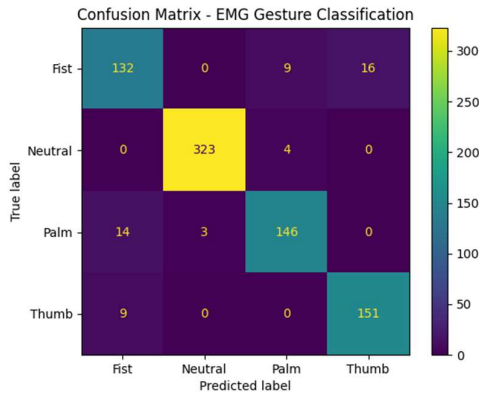


Figure 8. Confusion matrix

This behavior can be partly explained by the level of noise present in the signal, reflected in the SNR value obtained (around 10 dB), which limits the separability between classes.

Overall, the results indicate that, despite the simplicity of the single-channel system, it is possible to obtain competitive performance both in terms of morphological similarity and classification, which reinforces its viability for human-machine interaction applications.

4. Discussion

The results obtained highlight the relationship between the quality of the acquired EMG signal and the performance of the classification system. In particular, the SNR value obtained (around 10 dB) indicates the presence of a significant noise level in the signal, which directly affects the separability between classes. This effect is reflected in the confusion matrix, where classification errors can be observed between the fist and thumb gestures, as well as between the fist and palm gestures. These confusions can be explained by the similarity between certain muscle activation patterns, which, combined with the noise level present in the signal, makes their differentiation difficult for the classifier.

The palm gesture presents a lower correlation value than other gestures, which can be explained by its greater muscular complexity. Unlike gestures such as the fist or thumb, which involve more localized activations, hand opening involves a broader set of muscles, resulting in a more distributed activation pattern. The use of a single-channel system limits the capacity to capture all the spatial information of muscle activation, which results in lower morphological similarity with the NinaPro reference signal. This limitation is inherent to the acquisition system and highlights the importance of the number of channels in the characterization of complex gestures.

5. Conclusions and Future work

The developed system demonstrates the technical viability of low-cost single-channel electromyographic (sEMG) architectures for the discrimination of basic motor patterns, achieving high classification accuracy in simple gesture

recognition tasks. However, the experimental analysis highlights several critical limitations related to robustness, safety and temporal determinism of the system, which condition its application in real environments.

First, the signal-to-noise ratio (SNR) obtained of around 10 dB introduces statistical uncertainty that manifests in the overlap of classifier decision hyperplanes; specifically, the confusion matrix shows a critical error rate between the thumb and fist gestures, as well as between the fist and palm gestures, which would compromise actuator stability in the face of spurious false activations.

On the other hand, the temporal determinism of the system is penalized by the pre-processing and segmentation chain. The concatenation of 200 ms sliding windows, 50 ms moving average smoothing and the temporal consistency filter impose an intrinsic accumulated latency. This total delay of approximately 325 ms restricts the real-time response capacity of the closed-loop control.

Consequently, while single-channel myoelectric control is inadvisable for high-precision robotic applications or dynamic manipulation, its suitability is validated exclusively for discrete command interfaces, such as basic menu control or the activation and deactivation of simple devices (turning lights on and off, ventilation control...).

Finally, as future lines of work, it would be of interest to expand the number of recognized gestures, validate the system on a larger and more heterogeneous set of users and incorporate multi-channel architectures that allow improving separability between classes and increasing robustness against noise and physiological variations.

6. References

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